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جامعة خليفة
Khalifa University

Internal Audit Report
Review of Research Computing (RC)

Internal Audit Department
Khalifa University of Science and Technology
March - 2023

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March 2023

Steve Griffiths

Khalifa University of Science and Technology

We have completed our review of Research Computing. The review was conducted during January / February 2023 covering the period from January 2022 to December 2022.

The objective of our audit was to review the adequacy and effectiveness of internal controls system and procedures; and to ensure compliance with operating policies and procedures, laws and regulations.

We have discussed the contents of our report, including our recommendations, with the respective processes' owners and where relevant we have incorporated their comments and action plans to remediate our findings.

I would like to draw your attention to the issues highlighted in the executive summary, where opportunities for improvement were highlighted.

Yours Sincerely,

Abdulaziz Alkhoori

VP, Audit

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1. Executive Summary

1.1. Introduction

We have completed the Review of Research Computing (RC), in line with the Approved Internal Audit Plan and Scope Letter Dated January 2023. The Internal Audit Findings / Improvement Opportunities to address associated risks / control gaps were identified and reported as part of this report.

1.2. Scope of Work

The Scope of Work comprised of the following areas:

- Governance and Compliance.
- Research Computing Operations.
- Maintenance Operations.
- Procurement related Activities.

1.3. Audit Objectives

The Objective of this review is to provide independent and objective assessments of KU's business processes and systems of risk management, internal controls and governance processes. Further, assess compliance with policies/procedures and efficiency and effectiveness of operations and resource utilization.

1.4. Audit Process

The following are the key Audit Steps undertaken by the IA Department:

- Gain an understanding of the processes of the areas covered in the scope listed above.
- Reviewed adequacy and effectiveness of internal controls system and procedures.
- Reviewed compliance with the policies and procedures, laws and regulations.
- Agreed with process owners on action plans to improve control effectiveness / process performance.

1.5. Summary of Audit Findings

Controls need to be enhanced for **14%** of the observations. A balance of **50%** is within the moderate range and **36%** is within the low range and needs to be monitored and prevented from entering the high-risk category and moderate range as applicable.

Please refer to Appendix – 1 for detailed Ratings

The audit findings found during the review are summarized below. It should be noted that we have reported only on the exceptions detected during the course of our review.

1. Coverage of Privileged Access Management (PAM)

   Privileged Access Management (PAM) is an information security (infosec) mechanism that safeguards identities with special access or capabilities beyond regular users.

   However, the access of system administrators to RC Linux servers, are not covered through the Privileged Access Management (PAM) solution.

2. Backup Testing and Restore Process

   Backups play a critical role in protecting data from loss or corruption. In absence of adequate backups in place, RC will not be able to determine if data can be retrieved in a timely manner allowing researchers to continue their projects with minimal interruption.

   We noted lack of backup and restore test for RC resources resulting in potential loss of permanent data and impairment of operations.

2. Background

The Research Computing Department "RC" is responsible for ensuring the effective use of research computing resources. All users of Research Computing (RC) resources are responsible for complying with the conditions of use of RC resources and the applicable ICT acceptable use policies and guidelines. The Research Computing Committee (RCC) provides recommendations concerning resource allocation and priorities for RC resources.

- ### RC Services

RC supports researchers by understanding their computational research needs and recommending suitable applications and other research computing resources available at the KU. RC also assists researchers with grant proposals with a strong High-Performance Computing (HPC) component to describe the utilization of HPC resources in their research.

RC also ensures to resolve of faculty and student computing resource issues, provides programming consultancy, and code optimization, and resolves users' programming issues (based on Unix shell script, C/C++, Matlab, Python, MPI, OpenMP, CUDA). RC help build, install, maintain, and test scientific applications on the HPC clusters. Moreover, RC develops awareness and training programs for students, faculty, and staff with regard to research computing techniques, services, and facilities, including HPC application training to help users parallelize, debug and optimize their applications along with regular trainings workshops and tutorials, as well as one-on-one training for new users. In addition, RC performs required regular maintenance and upgrades incoordination with external vendors (Software) while hardware maintenance remains fully under Ankabut.

- ### Use of RC Resources

All research computing facilities, including HPC systems, operated and maintained by RC are accessible by authorized users and are subject to KU's policies - appropriate use of Information Technology. All users utilizing HPCs or other research computing facilities must acknowledge the use of facilities in their publications. Suitable text has been provided by RC and published and made available to users. Users are required to complete training courses prior to accessing RC resources.

- ### Resources Allocation

RC user group is created to educate and train users on the use RC services, challenges and formalize the Resources allocation plan. All users have access to a baseline level of service. Requests for additional resources are assessed by evaluating the need, availability, and impact on other users. This applies to all RC resources, whether storage, computing time or support. For demanding projects, the project funds can be utilized to purchase additional infrastructure/support for the project's exclusive use, or alternatively share the additional resource as appropriate.

- ### Scientific Computing Support

Scientific Computing Support is provided to users to aid in the effective provision and utilization of RC resources. The support provided will range from training, assisting researchers in running their codes on RC resources to providing consultation on performance analysis and code optimization.

- ### Committees

Research Computing Committee (RCC) has an approved charter with clear responsibilities. Nominations for RCC are made by Deans, with 2 faculty representatives from each college.

- ### Reporting

RC reports to Senior Vice President, R&D on a monthly basis addressing the utilization of resources for clusters and other administrative and operational matters.

3. Detailed Audit Findings

High	1. Coverage of Privileged Access Management (PAM)

Risk Ref : RC – 1

Observation/Process Enhancement

Privileged Access Management (PAM) is an information security (infosec) mechanism that safeguards identities with special access or capabilities beyond regular users. Like all other infosec solutions, PAM works through a combination of people, processes, and technologies.

Governance is a critical measure to limit and control Privileged Access Management (PAMs) to identify threats from any user with elevated admin privileges. Periodic privileged user access review is essential to verify that only legitimate users have admin access to applications or infrastructure.

During our review, we noted the access of system administrators to RC Linux servers, are not covered through the Privileged Access Management (PAM) solution.

Further, RC Linux servers are being accessed by end users with no privileged access.

Recommendation

- Access to RC Linux servers by system administrators should be configured through PAM solution.
- Procedure to perform periodic reviews of RC system administrator access and logged admin activities should be defined to ensure only authorized personnel have continued access and identify any anomalous activity.

Management Action Plan

Management Action Plan: Will liaise with Ankabut/IT to implement PAM for system administrators, and the process for monitoring. Worth noting that Ankabut has already implemented this for their VM's. Timeline has been decided after discussions with Ankabut.

Responsibility: Ankabut

Timelines: 31-Dec-2023

High	2. Backup Testing and Restore Process

Risk Ref : RC – 14

Observation/Process Enhancement

Backups play a critical role in protecting data from loss or corruption. In absence of adequate backups in place, RC will may be able to determine if data can be retrieved in a timely manner allowing researchers to continue their projects with minimal interruption.

During our review, we noted lack of backup and restore test for RC resources resulting in potential loss of permanent data and impairment of operations.

Recommendation

- Backup restore test and regular drill shall be executed to avoid loss of research data.
- Backup and restore drill shall be carried out in line with the approved DR Plan for KU

Management Action Plan

Management Action Plan: Backup service is run and maintained by Ankabut. Will discuss with Ankabut how regular backup and restore drills can be put in place. Timeline has been decided after discussions with Ankabut.

Responsibility: Ankabut

Timelines: 31-Dec-2023

Medium	3. Monitoring of Resources Allocation

Risk Ref : RC – 01, RC – 02

Observation/Process Enhancement

Low-overhead resource monitoring is a key to successful management of distributed high-performance computing environments, particularly when applications have well-defined quality of service (QoS) requirements. Resources utilization identifies the resources in use, users, and research projects utilizing such resources.

Currently, the resources are allocated via the workload manager, SLURM. The RC Department then uses SLURM and various custom tools, to monitor and report on the overall usage of the HPC cluster.

During our review, it was noted that the current monitoring mechanism is on a "need-to-know" Basis in absence of live monitoring tools in place resulting in retrospective monitoring over the resources to promote instant view over the availability of RC resources.

Recommendation

- Improve resource utilisation monitoring tools (live) and publish utilisation of RC resources in order to allow instant view over the availability of resources.
- Ensure instant reporting over the usage / availability of RC resources post implementation of live tools.

Management Action Plan

Management Action Plan: Will implement scripts/tools to monitor live utilisation of HPC resources and make this available to users.

Responsibility: Research Computing

Timelines: 30-May-2023

Medium	4. Monitoring of Research Software Licenses

Risk Ref : RC – 09

Observation/Process Enhancement

Software licenses serve several purposes, providing legal protection for software creators' intellectual property, generating revenue for software creators, controlling the use and customization of software, governing the use and distribution of open-source software, and ensuring compliance with legal and regulatory requirements.

During our review, it was noted that the monitoring over the usage of licenses related to research software and applications is being performed manually without taking into consideration applying a systemic approach to promote dynamic monitoring over licenses, for better optimization of licenses and subsequently, KU resources..

Recommendation

- Implement a systematic approach towards the monitoring of licenses.
- Ensure access is provided to authorized personnel only.

Management Action Plan

Management Action Plan: Software licenses and license servers are managed by Ankabut, and license usage monitoring needs to be done at the license server level. Will need to discuss with Ankabut regarding implementation and timelines.

Responsibility: Ankabut

Timelines: 31-Dec-2023

Medium	5. Documentation Requirements

Risk Ref : RC – 04

Observation/Process Enhancement

IT Documentation assists users understand software and hardware architecture, interdependencies, and how to use systems efficiently. Documentation plays a crucial role in helping students and researchers to effectively use software and GPU hardware, maximize their capabilities, and advance their research outcomes.

However, the set of RC documentation related to key applications are not updated / organized and lacks key details of the applications and tools being utilized by researchers on HPCs.

Recommendation

- Ensure documentation is updated to address the key applications.
- Reorganize documentation to assist users to navigate and locate the required information.

Management Action Plan

Management Action Plan: Will update and reorganize documentation. Have already identified the specific applications and pages to be added.

Responsibility: Research Computing

Timelines: 31-Dec-2023

Medium	6. Research Computing Committee

Risk Ref : RC – 05

Observation/Process Enhancement

During our review of operations undertaken by the Research Computing Committee (RCC) against the approved Charter, we noted the following:

- **No of meetings:** As per the approved Charter, RCC should meet at least 4 times a year. However, the RCC has not met during the period from Jan 2022 to Dec 2022.
- **Minutes Of Meetings:** MoM's are not signed or approved in absence of a process in place to follow up on the agreed action plans from previous MOMs.
- **Committee Reformation:** As per the approved Charter, the Committee should be reviewed by April 2020. However, is action to comply with the Charter is yet to be taken.

Recommendation

- Ensure compliance with RCC Charter.
- Committee Charter should be updated to reflect the changing requirements of RC.
- Adequate documentation shall be maintained. MoMs and action points shall be documented and followed up regularly.

Management Action Plan

Management Action Plan: Charter will be updated, and processes put in place to comply with MoMs requirements and meetings to be scheduled for all of 2023.

Responsibility: Research Computing

Timelines: 31-Dec-2023

Medium	7. Outreach of activities and raising awareness of RC services and capabilities

Risk Ref : RC – 04

Observation/Process Enhancement

Outreach activities and raising awareness of RC services and capabilities help to advance research outcomes and foster a collaborative and supportive research community.

RC Department needs to raise its profile and visibility across KU. It has been commented in the RC Survey responses, lack of adequate awareness of RC services and resources by researchers / faculty (not all researchers / faculty are aware of RC services and resources).

Recommendation

- Conduct outreach / awareness sessions for end users in KU community.
- Conduct technical workshops and outreach activities to engage with Faculty and research community.
- Conduct regular surveys (i.e. annual) to test awareness of KU community – RC services and resources.

Management Action Plan

Management Action Plan: Will considerably step-up outreach efforts. Plan to present at professional development week, conduct 3 outreach activities and conduct a survey.

Responsibility: Research Computing

Timelines: 31-Dec-2023

Medium	8. Server Patch Management

Risk Ref : RC – 03

Observation/Process Enhancement

Patch management is the process of distributing and applying updates to software. These patches are often necessary to correct errors (also referred to as "vulnerabilities" or "bugs") in the software.

Patch management requires staying current on available patches, deciding which patches are needed for specific software and devices, testing and ensuring appropriate installation and documentation process.

During our review, we noted the current patch management is limited to end user Linux VMs and is managed manually / excluded from IT's current patch management system.

Recommendation

- Define and document procedure for patching all RC Linux servers.
- Any exceptions in applying the security patches should be documented and reported to Ankabut (IT Security team).
- RC should regularly patch the machines to ensure stable and safe operation

Management Action Plan

Management Action Plan: Enhance, update format, and formalize existing process for patch management, as well as investigate and implement tools to help with this task. Will implement process and patch machines regularly, and documents any exceptions.

Responsibility: Research Computing

Timelines: 31-Aug-2023

Medium	9. Management of HPCME operations

Risk Ref : RC – 2

Observation/Process Enhancement

In July 2022, HPCME Company has been appointed to provide HPC support services including cluster management software and system configuration changes etc.

The RC sysadmins are responsible for the day-to-day operations related to HPC support with limited authority to perform the HPC configuration updates.

However, this arrangement has led to delays especially when updates to HPC configuration are needed resulting in full dependency on the response times / necessary actions provided by the Company to address such configurations.

Recommendation

- Better utilize in-house capabilities and Reduce dependency on external companies by taking full control over HPC configuration rights.
- Ensure compliance with contract terms and conditions. A detailed checklist can be developed to ensure all deliverables are made as per contract.

Management Action Plan

Management Action Plan: Will make arrangements to take control of configuration management from HPCME and use them in a support capacity. Will follow up regularly with HPCME regarding their SLA obligations.

Responsibility: Research Computing

Timelines: 30-Jun-2023

Low	10. Virtual Machines Resources

Risk Ref : RC – 08

Observation/Process Enhancement

A virtual machine environment is a software-based emulation of a physical computer system. It allows multiple virtual machines to run on a single physical machine, each with its own isolated operating system and set of resources, such as CPU, memory, and storage.

However, following the current practice adopted by RC, users can request Virtual Machines (VM's) via email communication without taking into consideration adopting an online mechanism / platform for the administration of such requests which may lead to inefficiencies / delays in providing the required resources in addition to the service / resources not being visible to all users.

Recommendation

- Create online application process for the receipt and processing of requests - VM resources.
- Awareness session shall also be provided to users for better utilisation of such online tools

Management Action Plan

Management Action Plan: Will create process and engage Ankabut Software development team to implement online form on the KU portal, similar to the HPC user and project creation process and form. Aim to have this ready to present at Professional Development week presentation in August, though this may depend on the Software Development teams' availability.

Responsibility: Research Computing

Timelines: 31-Dec-2023

Low	11. Project Management

Risk Ref : RC – 13

Observation/Process Enhancement

Project management is to plan and manage a project to successfully complete its listed goals and deliverables.

During our review, it was noted that the projects are managed individually via separate project plans with no holistic view over the projects and hence, lack of visibility of staff commitments.

Recommendation

- Improve existing project management mechanism to enable holistic overview of all the projects under RC.
- Assign prioritization / ranking mechanism of projects (i.e. major - minor/ high - low priority etc.) for better planning and execution of projects.

Management Action Plan

Management Action Plan: Aim to enhance processes and make use of tools to aid in overall visibility and ranking of projects.

Responsibility: Research Computing

Timelines: 30-Sep-2023

Low	12. Linkage between User Creation at KU Portal and User Setup at HPC System

Risk Ref : RC – 10

Observation/Process Enhancement

System integration between disconnected resources is a key for eliminating manual interaction among applications.

Our review showed that the user list in HPC System is not linked to KU Active Directory in order to determine the legitimacy of users of KU IT Systems.

Recommendation

- Ensure only authorized personnel are accessing VMs and HPCs.
- Engage with Ankabut department to evaluate the feasibility and timeline for integrating these systems.

Management Action Plan

Management Action Plan: Will create scripts/processes to check that all users with access to RC resources are authorized and still at KU. Will engage with Ankabut to investigate how this can be automated, and feasibility of integrating with existing KU systems, processes, and mechanisms.

Responsibility: Research Computing

Timelines: 30-Sep-2023

Low	13. Standards Procedures

Risk Ref : RC – 15

Observation/Process Enhancement

Standard Procedures are crucial to specify roles, responsibilities, and authorized personnel to execute and follow up on various aspects of RC operations.

The existing procedures are not consistent and detailed in addressing the following areas (not limited to):

- HPC Patches update procedures.
- HPC Operating system update procedures.
- Backup Process and Procedure.

Recommendation

- Update standard procedure in consistent and standard format, covering the areas highlighted in the audit finding.
- Ensure adequate implementation of procedure and awareness thereof.

Management Action Plan

Management Action Plan: Will update all procedures to a standardised format, and improve the documentation

Responsibility: Research Computing

Timelines: 31-Oct-2023

Low	14. Planned vs. actual Manpower

Risk Ref : RC – 01

Observation/Process Enhancement

According to the approved Organization Chart, Research Computing consists of 4 staff members out of a total 6 planned as per the approved chart (**67%**).

It is worthwhile to mention that availability of resources will enhance the expansion of RC operations

Recommendation

Vacancies should be filled in line with the approved Organization Chart.

Management Action Plan

Management Action Plan: In the process in recruiting for the vacant posts. Aim is to have both filled by Q3, however this will depend greatly on the availability and ability to attract suitable candidates.

Responsibility: Research Computing

Timelines: 31-Nov-2023

Appendix

Findings Rating

The audit findings have been grouped together in a logical manner to appropriately convey the risk and impact. The audit findings have been categorized based on their criticality rating as **High**, **Medium** and **Low**. The definitions of the findings criticality / distribution are provided in the table bel

	Finding Rating	Definition	Number of Findings	Total
●	High	Finding is serious and deserves immediate attention by the Process Manager. The finding is reported to the Audit Committee quarterly.	2	14
●	Medium	Finding impacts the accomplishment of the Process objectives. Corrective action is required by management. If not resolved, the finding could result in inefficient use of entity resources and or potentially disrupt business processes.	7	
●	Low	Finding to be reported to Management but is of a minor risk to the entity. Findings will not be reported to the Audit Committee unless the finding remains open after the follow up audit.	5	